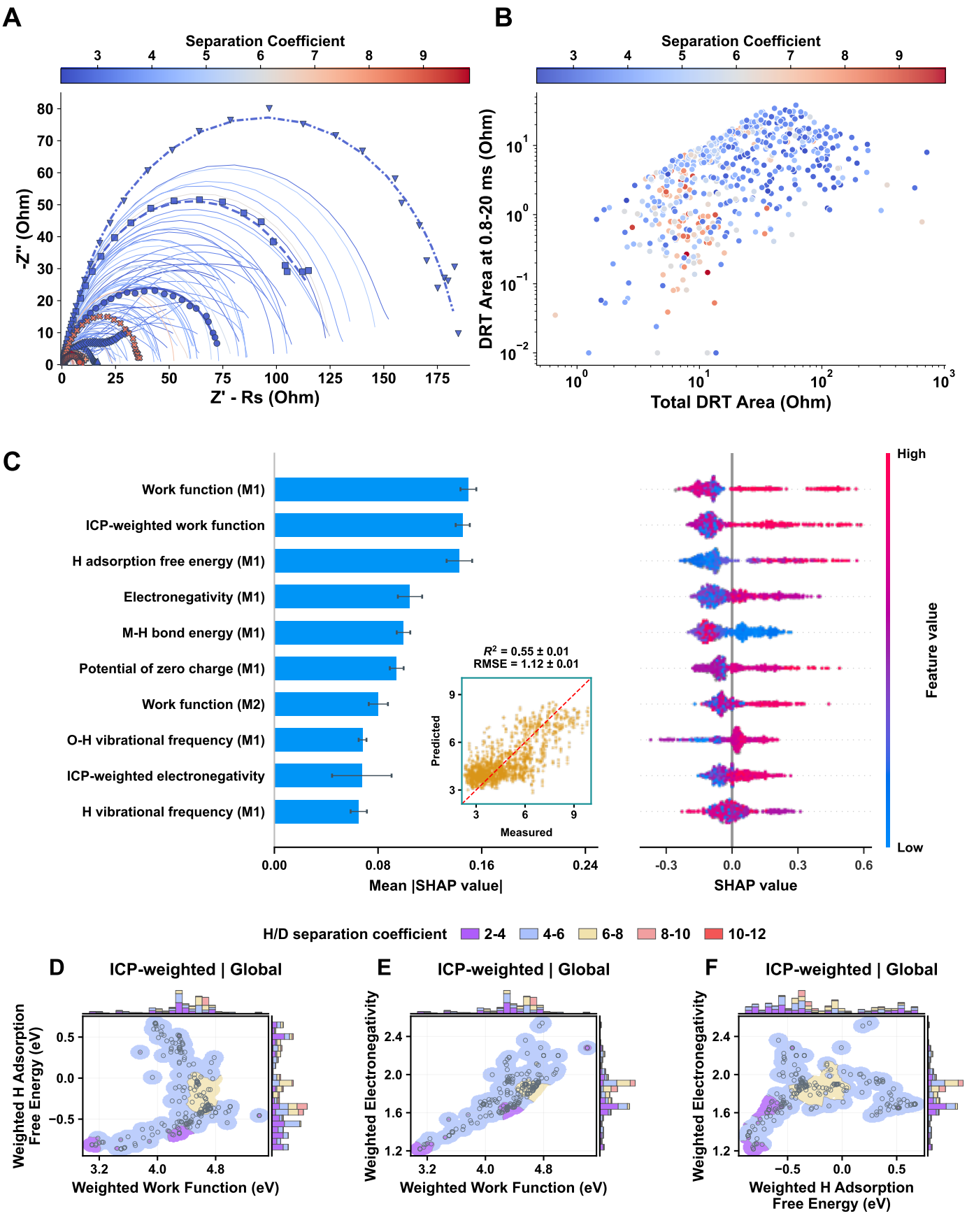

A
Separation Coefficient
3 4 5 6 7 8 9
-Z'' (Ohm)
Z' - Rs (Ohm)
B
Separation Coefficient
3 4 5 6 7 8 9
DRT Area at 0.8-20 ms (Ohm)
Total DRT Area (Ohm)
C
Work function (M1)
ICP-weighted work function
H adsorption free energy (M1)
Electronegativity (M1)
M-H bond energy (M1)
Potential of zero charge (M1)
Work function (M2)
O-H vibrational frequency (M1)
ICP-weighted electronegativity
H vibrational frequency (M1)
Mean |SHAP value|
R² = 0.55 ± 0.01
RMSE = 1.12 ± 0.01
Predicted
Measured
SHAP value
High
Feature value
Low
H/D separation coefficient
2-4
4-6
6-8
8-10
10-12
D
ICP-weighted | Global
Weighted H Adsorption Free Energy (eV)
Weighted Work Function (eV)
E
ICP-weighted | Global
Weighted Electronegativity
Weighted Work Function (eV)
F
ICP-weighted | Global
Weighted Electronegativity
Weighted H Adsorption Free Energy (eV)